# Xinlu Liu

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Algorithm engineer at Tencent Music Entertainment, with a **master's degree in Computer Science and Technology from Fudan University**. My interests include AI-assisted music production workflows, audio-effect representation learning, and real-time music systems.

## Experience

**Tencent Music Entertainment (TME), Galaxy Audio Effect Team**
Algorithm Engineer — Mar. 2024 – Present

**Selected Projects**

**RelFx: Relative Audio Effect Representation** — Mar.–Jun. 2026

- Proposed a model that learns changes in audio effects from ordered audio pairs, using contrastive distance learning and cross-segment sampling. Representation training requires no dry references.
- Reduced MR-STFT error by **13.3%** versus Fx-Encoder++ in MUSDB18 effect-transfer evaluation (1.601 to 1.388). **First author, ISMIR 2026**; released code and a model checkpoint.

**AutoMix: Automatic Song Transitions** — 2025–2026

- Led development of the automatic song-transition framework, combining rhythm, structure, and vocal cues to choose transition points and strategies; **shipped two versions of AutoMix in QQ Music**.

**IntelligentFX: Music-aware Audio Effects** — Jun.–Dec. 2024

- Led the design of music-aware adaptive audio processing for whole songs and individual sections.
- Drove product integration and the **launch of IntelligentFX in QQ Music 14.0**.

**Music Therapy** — Jun.–Sep. 2023; Mar.–Jun. 2024

- Led core algorithm and system development for personalized music generation supporting sleep, focus, and relaxation.
- **Launched in QQ Music** and **integrated into the soundcore app for sleep headphones** through a hardware partnership.

**Other Systems**

- **SoundDrive:** Led interaction design, prototyping, system development, and vehicle integration for an adaptive music system **deployed in ZEEKR 001 vehicles**, with driving signals controlling effects across 8 stems.
- **Natural-language audio tuning:** Built a prototype for conversational effect editing and before/after listening, connecting LLM tool calls to DSP parameters and manual controls.

## Selected Publications

**Xinlu Liu**, H. Lin, W. Wei, Z. Yan. *Beyond Dry References: Learning Relative Audio Effects Representations via Contrastive Distance Learning.* — ISMIR 2026
First author; accepted. Paper · Project

**Xinlu Liu**, et al. *LC-Beating: an online system for beat and downbeat tracking using latency-controlled mechanism.* — ICME 2023
First author; oral presentation. Paper · Code

J. Qian, **X. Liu**, et al. *Stripe-Transformer: deep stripe feature learning for music source separation.* — EURASIP JASMP 2023
Second author. Paper

Y. Wu, **X. Liu**, et al. *XBeat: A hybrid CNN-Transformer model for beat and downbeat tracking.* — CSMT 2023
Second author. Paper

## Ongoing Research

- **Post-hoc FX Registration:** Developing reference-guided audio-effect transfer with frozen RelFx features, parameter prediction for registered effects, and candidate rendering and ranking. Manuscript in preparation.

## Education

**Fudan University** Sep. 2021 – Jan. 2024
M.Eng. in Computer Science and Technology Shanghai, China
Audio and Music Technology Laboratory (FD-LAMT). Research in beat/downbeat tracking and music source separation.

**Huazhong University of Science and Technology** Sep. 2017 – Jun. 2021
B.Eng. in Automation Wuhan, China

## Technical Skills

- **ML and programming:** Python, PyTorch, C++.
- **Audio engineering:** Real-time DSP, audio SDK development, effect chains, and audio tool integration.